

Obs
adverse

Categorical: IFR-or-worse event	
Hit	Miss
False alarm	Correct neg

Forecast adverse →

→ **HSS, POD, FAR, CSI**

Probabilistic

```
adverse_probability(hour):  
    prevailing adverse → 1.0  
    PROB30 → 0.3 · PROB40/TEMPO → 0.4  
    else 0.0
```

Brier = mean (p – event)²

→ **BSS = 1 – BS/BS_clim**

Weighted per-hour score

prevailing exact → 3.0

TEMPO captures obs → 2.0

PROB≥40 captures → 1.5

PROB/off-by-one → 1.0

else → 0.0

→ **mean weighted score**

Promotion target: HSS (categorical, primary) with BSS (probabilistic); element MAE + weighted score are diagnostics